Comparative Assessment of Advanced Water and Wastewater Treatment Technologies

Treatment Performance, Energy Consumption, and Operational Limitations

INDEPENDENT RESEARCH PROJECT

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Abstract

This project presents a focused comparative assessment of eleven selected peer-reviewed studies on advanced and established water and wastewater treatment approaches. The selected literature covers membrane-based processes, advanced oxidation processes (AOPs), plasma-based treatment, and studies combining multiple treatment or assessment approaches. Information on wastewater matrices, target pollutants, treatment scale, reported performance, energy indicators, costs where available, and operational limitations was extracted into a structured Excel dataset. The studies report high treatment performance in several applications; however, the reported values describe different pollutants, matrices, treatment objectives, and operating conditions and therefore do not support a universal technology ranking. Energy metrics are also heterogeneous, including specific energy consumption, energy per order, energy yield, discharge power, annual plant energy use, and life-cycle indicators. Accordingly, numerical comparisons are retained only within individual studies or where the reported basis is sufficiently aligned. The assessment highlights the importance of operational reliability, fouling control, chemical requirements, energy demand, scale-up, and system boundaries when considering the practical suitability and sustainability of treatment technologies.

1. Introduction

Water and wastewater treatment supports public health, ecosystem protection, and the sustainable use of water resources. Municipal and industrial wastewaters may contain conventional organic matter and nutrients as well as dyes, pesticides, pharmaceuticals, and other micropollutants. Treatment selection therefore requires attention to both the contaminant characteristics and the performance and operating requirements of the process.

Membrane-based processes separate contaminants through selective barriers and can produce high-quality effluent, but their performance and cost may be affected by fouling, pressure demand, cleaning requirements, and aeration. Advanced oxidation processes generate reactive species that can transform persistent organic contaminants; their effectiveness may depend on pH, oxidant dose, mass transfer, and radical-scavenging constituents. Plasma-based systems can generate reactive oxygen and nitrogen species and may be combined with ozone or adsorptive media, although reactor configuration, energy use, and scale-up remain important considerations.

A direct comparison across these technology families is challenging. The selected studies investigate different wastewater matrices, target pollutants, scales, treatment objectives, and energy indicators. A removal percentage for a dye, an energy-efficiency score for a wastewater treatment plant, and an electrical energy per order (EE/O) value do not represent the same outcome. This project therefore adopts a context-sensitive comparative approach rather than calculating broad category averages or ranking technologies using non-equivalent metrics.

The aim is to assess the treatment performance, reported energy requirements, and operational limitations documented in eleven selected studies, and to identify which comparisons are meaningful within the evidence available.

2. Methodology

2.1 Research approach and scope

The work was conducted as a focused comparative literature assessment. It is not presented as a systematic review or meta-analysis. The scope was limited to the eleven studies already selected for the project; no additional papers were introduced during the audit or finalization.

2.2 Literature selection

Google Scholar was used as the principal source for identifying relevant publications. The selected studies were retained because they addressed water or wastewater treatment technologies, technology comparisons, energy-related performance, or operational and sustainability considerations. A complete search log, screening flow diagram, search dates, search strings, and counts of excluded records were not maintained. The selection should therefore be understood as criteria-based and focused, not exhaustive.

2.3 Data extraction

Information was organized in the accompanying Excel workbook. Extracted fields include paper identifier, full title, publication year, treatment technology, descriptive technology category, wastewater type, target pollutant, study scale or assessment type, reported treatment performance, operating conditions, treatment time, reported energy value and unit, energy-efficiency indicator, treatment cost where available, operational limitations, sustainability findings, and source citation. Where a study reported multiple processes or scenarios, values were retained with their study context rather than collapsed into a single value.

2.4 Technology classification

The eleven studies were assigned to five descriptive categories: Membrane, AOP, Plasma, Membrane / Other, and Other. The categories describe the composition of this selected dataset. They do not imply that all studies within a category used equivalent processes or that the category counts represent the prevalence of technologies in the wider research literature.

2.5 Performance and energy interpretation

Reported performance indicators were preserved in their original form. These include pollutant-removal percentages, decolorization, mineralization, effluent concentrations, water recovery, and plant-level energy-efficiency scores. These indicators were not treated as interchangeable. In particular, the StoNED score reported for wastewater treatment plants is an energy-efficiency benchmark, not a removal percentage.

Energy indicators were also kept distinct. The dataset includes values expressed as kWh/m³, J/m³, W, g/kWh, kWh/m³/order (EE/O), kWh/kg pollutant, and annual MWh/year. Even where units appear similar, differences in denominators, process boundaries, treatment objectives, and system scale may prevent direct comparison. No cross-category mean removal efficiency, mean energy consumption, or overall technology ranking is reported.

2.6 Operational and sustainability assessment

Operational limitations were extracted from the selected papers and synthesized qualitatively. The assessment considers membrane fouling, pressure and aeration demand, chemical cleaning, pH adjustment, oxidant dosing, electrode stability, ozone transfer, radical scavenging, loading and capacity utilization, maintenance, mechanical reliability, and scale-up. Sustainability findings are interpreted in the context of each paper's functional unit, cost basis, and life-cycle system boundary.

3. Results and Discussion

3.1 Composition of the selected literature

The eleven studies comprise five membrane-category papers, two AOP papers, two plasma papers, one Membrane / Other paper, and one Other paper. The largest group is therefore membrane-based, but this reflects the composition of the selected set rather than evidence that membrane technologies are more common or more effective overall.

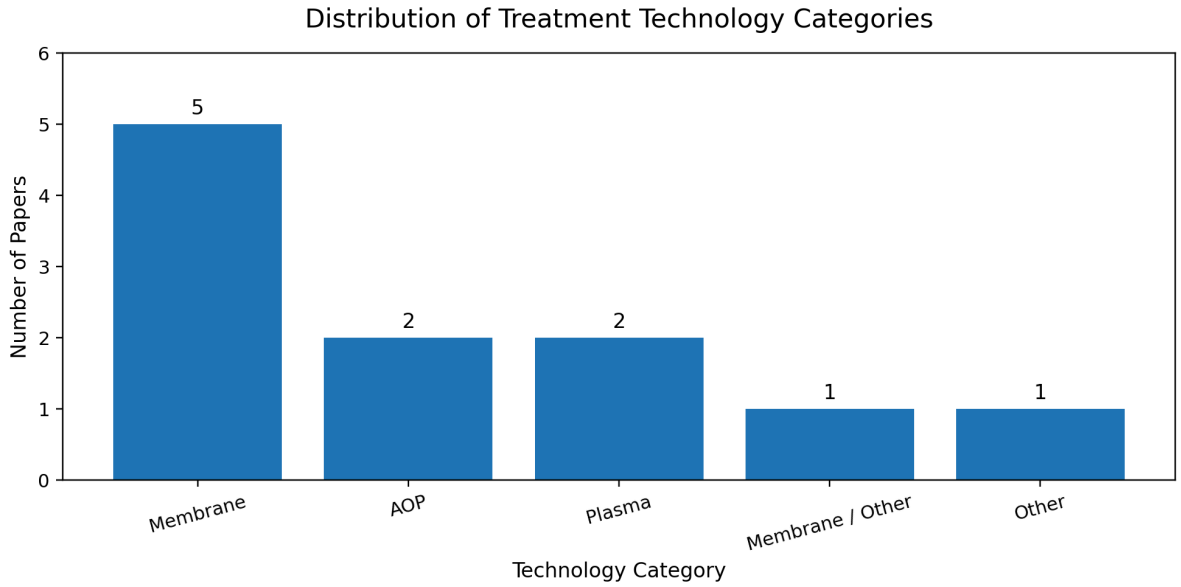


Figure 1. Distribution of treatment technology categories among the eleven selected studies.

3.2 Reported treatment performance

Several studies reported high treatment performance within their own experimental or operational settings. In the Ethiopian case-study assessment by Tizazu et al. (2026), the membrane bioreactor achieved 98.94% COD and 99.12% TSS removal, while the AANAF sequencing batch reactor achieved 99.78% COD and 99.66% BOD removal. The paper also describes stable performance and low energy demand for waste stabilisation ponds (WSPs) operated near design capacity. No WSP pollutant-specific removal percentage is reproduced here because the extracted evidence did not support the percentage previously associated with that entry. The study also demonstrates that low hydraulic-capacity utilization can increase unit treatment costs even where removal performance is high.

For electrochemical oxidation of landfill leachate, Asaithambi et al. (2020) reported 100% colour removal and 97% COD removal using a photo-electro-Fenton process, with a reported energy

consumption of 3.10 kWh/m³. The result is specific to the studied leachate matrix and operating conditions, including acidic pH and hydrogen peroxide dosing.

Shafei et al. (2026) reported decolorization efficiencies of 97.61% for methyl orange and 95.78% for crystal violet using a multi-faceted rotating arc discharge plasma reactor. Vanraes et al. (2017) reported pesticide-specific removal values ranging from 64.8% to 97.0% in a falling-water-film dielectric barrier discharge system combined with activated-carbon textile and ozone. These percentages relate to different pollutants and matrices and should not be interpreted as a head-to-head ranking.

Wardenier et al. (2019) compared ozone-, UV-, and plasma-ozonation-based processes for four micropollutants in a synthetic water matrix. Plasma-ozonation achieved more than 99.8% removal for three target compounds and 95.5% removal for atrazine after 20 minutes under the reported conditions. This is a useful within-study comparison because the alternatives were assessed in a shared experimental framework.

Molinos-Senante and Maziotis (2022) assessed 203 Chilean wastewater treatment plants using a StoNED energy-efficiency score, reporting a mean score of 0.433. This score describes plant-level energy-efficiency performance and is not a pollutant-removal percentage. The paper's reported average potential energy savings of 203,413 MWh/year is retained as a study-level figure without assigning it to an individual plant.

3.3 Energy-related findings

Energy findings are most informative when interpreted within the system in which they were measured or modelled. Alsarayreh et al. (2020) reported that an energy-recovery device reduced energy consumption by approximately 47–53.8% in a medium-scale brackish-water reverse-osmosis plant; the reported specific energy consumption reached 0.786 kWh/m³ at 30 °C with 90% energy-recovery-device efficiency. This is a within-system comparison and should not be ranked directly against laboratory oxidation processes or municipal wastewater plants.

Lin et al. (2026) compared parallel full-scale vibrating and aerated membrane bioreactor trains over approximately 800 days. The reported total specific energy consumption was 0.34 ± 0.02 kWh/m³ for the vibrating configuration and 0.45 ± 0.04 kWh/m³ for the aerated configuration, corresponding to an approximately 24% reduction. The energy used specifically for membrane fouling control was reported to decrease by 75%. These are distinct indicators: the 75% value applies to fouling-control energy, not total plant energy.

Wang et al. (2021) compared osmotically assisted reverse osmosis (OARO) and low-salt-rejection reverse osmosis (LSRRO) through process modelling. The preferred configuration depended on feed salinity: LSRRO had lower specific energy consumption under the lower-salinity scenarios, whereas OARO performed better in the higher-salinity scenarios. Because this is a modelling comparison, the results should not be described as measured full-scale plant performance.

Other studies report energy in J/m³, discharge power, energy yield in g/kWh, EE/O, or annual MWh/year. These metrics are retained separately in the Excel dataset. Converting units alone would not resolve differences in system boundaries, pollutant targets, treatment scale, or indicator definitions. Accordingly, no category-level average energy value is calculated.

3.4 Operational limitations of AOP and plasma-based processes

The photo-electro-Fenton study by Asaithambi et al. (2020) used acidic conditions (pH 3) and hydrogen peroxide dosing. Electrode passivation under direct-current operation was identified as an operational

concern, while alternating-current operation helped mitigate this issue under the conditions studied. These requirements influence chemical use, maintenance, and process control.

Mass transfer and radical-scavenging reactions can constrain ozone- and plasma-assisted oxidation. Wardenier et al. (2019) reported approximately 44% ozone uptake in the investigated plasma-ozonation setup, indicating that a substantial fraction of supplied ozone was not transferred to the liquid phase. Phosphate buffer constituents and excessive hydrogen peroxide can also consume reactive species and reduce oxidation efficiency. Vanraes et al. (2017) further showed that pesticide structure and the presence of inorganic constituents influence removal and energy performance in plasma-based treatment.

Taken together, these studies indicate that the practical performance of AOP and plasma systems depends on more than oxidation potential. Chemical dose, pH, electrode condition, gas-liquid mass transfer, scavenging constituents, reactor configuration, and the formation or persistence of transformation products all require consideration.

3.5 Operational limitations of membrane technologies

Membrane systems can provide effective separation and high-quality effluent, but fouling and scaling can reduce permeability and flux, increase pressure requirements, and necessitate physical or chemical cleaning. These activities contribute to energy use, maintenance needs, and operating costs.

Wang et al. (2020), assessing full-scale membrane bioreactor plants in the East Taihu basin, reported that energy consumption accounted for 48.09% of operating expenditure. The study also identified the importance of influent carbon availability, chemical dosing, membrane fouling, and the broader operating context when assessing MBR sustainability.

Haider et al. (2026) compared a decentralized low-cost membrane bioreactor scenario with a centralized extended-aeration activated-sludge scenario using life-cycle assessment. The findings depend on the selected functional unit and system boundary, including the decentralized-versus-centralized service arrangement. Consequently, its life-cycle impact results should be interpreted separately from operational SEC values reported by other papers.

Lin et al. (2026) demonstrated improved fouling resistance and lower energy use for the vibrating membrane configuration under the full-scale conditions investigated. However, long-term mechanical reliability of vibration-drive assemblies remains a practical consideration for wider deployment. Wang et al. (2021) also showed that membrane configuration and feed salinity influence energy requirements in high-salinity brine management.

3.6 Comparative synthesis

The selected papers illustrate complementary rather than interchangeable treatment functions. Membrane processes are relevant to separation and high-quality effluent production; AOP and plasma-based approaches are relevant to oxidation and transformation of persistent organic contaminants; and plant-wide energy and life-cycle studies provide broader operational or sustainability context. The most defensible direct comparisons are those conducted within an individual study under a shared experimental, operational, or modelling framework, such as VMBR versus AMBR, OARO versus LSRRO across defined salinity scenarios, and the AOP alternatives assessed by Wardenier et al.

Cross-study comparisons require caution because pollutant type, wastewater matrix, scale, operating conditions, and energy indicator vary. For example, EE/O cannot be treated as interchangeable with SEC, and a StoNED score cannot be plotted as a removal percentage. Technology selection should

therefore consider the treatment objective, contaminant characteristics, operating stability, chemical and energy demand, maintenance, cost boundary, and scale-up requirements together.

4. Limitations of the Assessment

- The dataset contains eleven selected papers and is not an exhaustive representation of water and wastewater treatment research. The small and uneven number of studies in each category limits generalisation.
- The literature selection was criteria-based, but a complete search log, exact search dates and strings, screening counts, and exclusion records were not maintained. The work is therefore not a systematic review.
- The studies differ in wastewater matrices, target pollutants, treatment objectives, scale, operating conditions, and reported indicators. These differences prevent broad statistical pooling and universal technology rankings.
- Some papers report several process configurations or scenarios, while others assess whole treatment plants or life-cycle impacts. A single category label cannot capture every process detail or system boundary.
- Cost and sustainability data are not consistently available or reported using aligned currencies, reference years, functional units, and system boundaries. Qualitative synthesis is therefore more appropriate for several comparisons.
- Reported results are based on the information available in the selected publications and the extracted dataset. Missing or ambiguous values have not been estimated; uncertain interpretations are explicitly qualified.

5. Conclusion

This comparative assessment examined eleven selected studies covering membrane-based treatment, advanced oxidation, plasma-based treatment, and broader plant-level or combined technology assessments. The literature reports strong treatment performance in several applications, including organic matter, dyes, pesticides, and micropollutants. However, the results were obtained under different matrices, scales, and operating conditions and do not support a universal ranking of treatment technologies.

Operational constraints are central to practical evaluation. AOP and plasma systems may require careful control of pH, oxidant dosage, electrode condition, ozone transfer, and radical scavenging. Membrane systems may be constrained by fouling, pressure and aeration demand, chemical cleaning, and long-term mechanical reliability. Plant-level energy-efficiency and LCA studies further show that loading, process configuration, service arrangements, and system boundaries can materially affect sustainability conclusions.

The most reliable comparisons are those made within a common study framework and using clearly defined indicators. Future comparative work would benefit from consistent reporting of energy metrics, functional units, treatment boundaries, operating conditions, and long-term operational data. Such consistency would strengthen the evidence base for selecting treatment technologies under specific wastewater and sustainability requirements.

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Appendix A. Study-level evidence and comparison notes

Paper	Category	Study / treatment focus	Study type / scale	Key interpretation note
1	Membrane / Other	MBR, WSP, UASB+TF, SBR and DAF case studies	Full-scale case studies	Within-paper comparison only; no WSP removal percentage inferred.
2	Plasma	Multi-faceted rotating arc discharge plasma for textile dyes	Laboratory	Reports dye decolorization, discharge power, energy cost and g/kWh; do not merge these metrics.
3	Other	Energy-efficiency benchmarking of 203 Chilean WWTPs	Full-scale plant dataset	StoNED score is an energy-efficiency benchmark, not removal efficiency.

Paper	Category	Study / treatment focus	Study type / scale	Key interpretation note
4	Membrane	Brackish-water RO with energy-recovery device	Medium-scale operational plant	ERD comparison is meaningful within the plant/system boundary; publication year 2020.
5	AOP	Photo-electro-Fenton treatment of landfill leachate	Laboratory	Acidic pH, peroxide dosing and electrode behaviour affect operation.
6	Membrane	Decentralized LC-MBR versus centralized EA-ASP life-cycle assessment	Operational case / LCA scenario	Interpret LCA results using the study's functional unit and system boundary.
7	Membrane	Vibrating versus aerated membrane bioreactor	Parallel full-scale operation	Keep total SEC reduction separate from fouling-control energy reduction.
8	Plasma	Falling-water-film DBD plasma, activated-carbon textile and ozone	Laboratory	EE/O and G50 are distinct energy indicators; pesticide-specific performance varies.
9	Membrane	Full-scale MBR wastewater treatment plants in East Taihu basin	Full-scale multi-plant assessment	Energy, OPEX, fouling and influent characteristics must be considered together.
10	Membrane	OARO versus LSRRO for brine management	Process modelling / scenario analysis	Direct comparison within the modelled salinity scenarios; not full-scale plant data.
11	AOP	Ozone, UV, peroxide and plasma-ozonation AOP comparison	Laboratory	Within-study AOP comparison; EE/O and projected ozone-transfer case must remain distinct.

Note: The accompanying Excel workbook contains the full extracted fields, comparability matrix, analysis rules, reference register, technology-category distribution chart, and audit log. All quantitative values should be interpreted with the source paper's original definitions and operating context.